

[WORKING TITLE — see PROMPT 19 E2]

On the Evaluation of Structural Priors in Unrolled Channel Estimation for Rydberg Atomic Receivers

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Abstract—[ABSTRACT IS A STUB — PROMPT 19 E2.] It cannot be written until batch C2 settles which of the three P29 outcomes fires, because the three are three different papers and the abstract is the sentence that commits to one. What it must contain either way: the gradient-concentration finding (89.7% of the gradient below 5 dB, a per-bin share spanning a factor of 30.0); the matched-adequacy protocol; the corrected loss and its +2.171 dB three-seed gain at $\text{SNR} \geq 5$ dB with no low-SNR cost; the pilot-efficiency distinction at 2.231 dB; and simulation-only scope. What it must not contain until C2 reports: any scalar for the structural gain.

Index Terms—Algorithm unrolling, channel estimation, evaluation methodology, phase retrieval, Rydberg atomic receiver, structural priors.

I. INTRODUCTION

UNROLLING an iterative estimator into a trained network [1], [2] has become standard for channel estimation, including for Rydberg atomic receivers, where the magnitude-only readout makes the problem one of phase retrieval [3], [4]. A recurring next step is to add an explicit structural prior — a low-rank projection, a sparsity constraint — inside the unrolled loop, and to report the resulting improvement as evidence that the prior supplies information the network lacks.

This paper argues that such evidence is usually not decisive, and that the reason is a convention so widespread it is rarely stated: the training loss is a per-sample normalized error, averaged over a training set spanning a wide SNR range.

This is a finding about evaluation practice, and we did not arrive at it as critics. We built a Hankel-structured unrolled estimator, measured an advantage at high SNR, and believed it. The control described in Sec. III was run to characterize that advantage, not to remove it. The first casualty of the protocol we propose was our own method, and we report it because a result that survives an author's attempt to keep it is worth more than one that was never tested.

Related work on SNR-dependent loss design. The imbalance itself is known and we do not claim it. Wiesmayr *et al.* [5] state it directly for ML-assisted detection and decoding: a system learns one parameter set while operating over a range of SNRs, the training data is sampled from that range, and the aggregate loss is then dominated by the low-SNR samples, because a small relative improvement at low SNR moves the

cost further than a large one at high SNR. They propose SNR deweighting as the remedy. Work on deep OFDM channel estimation has separately examined training on SNR-restricted samples, reporting the expected trade of low-noise accuracy against high-noise accuracy.[TODO: VERIFY CITATION: arXiv:2401.05436, fields unconfirmed] What we take from that line of work is the mechanism and the remedy. What is not taken from it, and what this paper is about, is the consequence for *attribution*: that a known training defect manufactures apparent evidence for a structural prior, so that crediting the prior without controlling for the defect is unsound. Wiesmayr *et al.* identify the pathology and fix it; they do not ask what it does to the evidence.

Contributions.

- 1) An attribution decomposition of an unrolled estimator's gain, separating the learned filter, the attention block, and unrolling itself against a matched non-unrolled control.
- 2) The consequence of a known training pathology for *attribution*: per-sample normalized error over mixed SNR concentrates the gradient at low SNR [5], so any mechanism that improves high-SNR behaviour will appear to add information, and a structural prior credited on that evidence is credited for the training's failure rather than its own contribution. We give a matched-adequacy protocol that separates the two.
- 3) Two properties of the structural contrast that a scalar hides: a per-bin sign pattern that replicates across training designs and seeds, and a dependence of the reported sign on the averaging window.
- 4) Confirmation that the known remedy [5] transfers to unrolled magnitude-only estimation, gaining +2.171 dB at $\text{SNR} \geq 5$ dB over three seeds — with a gain in every bin of every seed, including the two bins the weighting penalizes most.
- 5) A demonstration that pilot efficiency and pilot-count generalization are distinct and are conflated by a single reported curve.
- 6) [STUB] The structural-prior contribution, pending C2.

[STUB — INTRODUCTION, structural-prior claim]

One paragraph stating what the control did to Δ_H , and the corresponding contribution item. It cannot be written until batch C2 reports, because the three P29 outcomes give three different claims: a stable gain whose value depends on training adequacy; a quantity with no stable sign under any design; or

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a stable gain materially smaller than first reported. What is already established and belongs here whichever fires: under focused training Δ_H has no stable sign across three seeds, and the per-bin ordering — a cost at low SNR, a benefit at high SNR — replicates on every arm measured while the SNR at which it crosses does not.

II. SYSTEM MODEL AND ESTIMATOR

An N -element uniform linear array serves K users over P pilots. With $\mathbf{G} \in \mathbb{C}^{N \times K}$, pilots $\mathbf{S}$, known reference beam $\mathbf{B}$ and noise $\mathbf{W}$, the receiver observes magnitudes only,

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|, \quad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2). \quad (1)$$

Each user's channel is a sum of L_k plane waves on the array manifold, with $L_k \sim \mathcal{U}\{3, 7\}$ [6] and angles drawn uniformly on $(-90^\circ, 90^\circ)$ (Table I); this is a geometric ULA model, not a clustered one. The classical baseline is an expectation-maximization Gerchberg–Saxton iteration (EM-GS) [7]; the unrolled estimator (“URformer”) replaces T iterations with T learned layers, each a data-consistency step followed by a learned filter, a gate, and a Transformer block across users [3].

We implement this estimator faithfully on the geometric ULA model of (1) rather than reproducing any published result; the operating point differs. Nothing here should be read as a reproduction or a validation of prior work, nor as a critique of any specific paper. The convention we examine is near-universal, and our own estimator uses it.

The reference curve is genie-aided. Where a reference appears, it is an unstructured least-squares estimate formed from the *true* phase of the noiseless field. It is not a bound: it presumes information no receiver has, and being restricted to unstructured least squares it can be passed by an estimator that exploits channel structure — which we in fact observe below 5 dB in Sec. IV-A.

III. ATTRIBUTION AND THE MATCHED-ADEQUACY PROTOCOL

A. Where the gain comes from

Fig. 1 decomposes the unrolled estimator's 3.345 dB improvement over EM-GS. The learned per-layer filter, 980 parameters, contributes 0.147 dB; the attention block contributes the remaining 3.198 dB, or 96%. The filter's contribution does not grow with data — it falls slightly, $0.193 \rightarrow 0.147$ dB from 20k to 80k training samples — which is what a never-capacity-limited 980-parameter component should do.

Unrolling is doing real work independently of attention. Against a matched control that applies one Transformer block to a converged EM-GS estimate rather than unrolling, the unrolled estimator gains 0.920 dB, CI $[+0.888, +0.988]$, winning on 86.5% of trials.

B. The confound

The mechanism in this subsection is the one stated in [5]; what is new here is its magnitude in an unrolled magnitude-only estimator, and what it does to attribution.

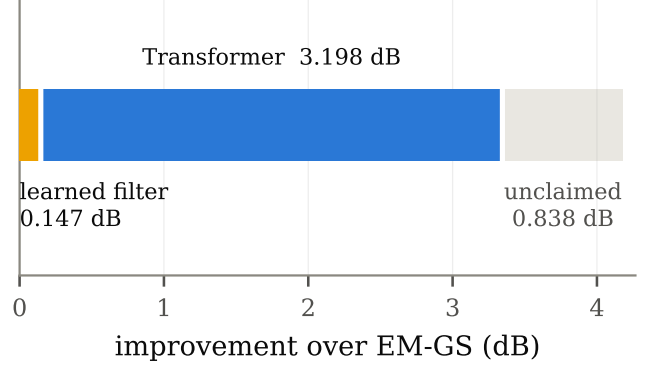


Fig. 1. Attribution of the unrolled estimator's gain over EM-GS at 80k training samples. “Unclaimed” is the remainder of the genie-aided least-squares headroom, which is a reference and not a bound.

The training loss is the per-sample normalized error $\|\hat{\mathbf{G}} - \mathbf{G}\|_F^2 / \|\mathbf{G}\|_F^2$, averaged over a batch drawn across the whole SNR range. Because the normalized error of a low-SNR realization is orders of magnitude larger than that of a high-SNR one, this average is dominated by the low-SNR tail. Measured directly on our estimator, 89.7% of the gradient comes from trials below 5 dB, and the per-bin gradient share spans a factor of 30.0 across the range (Fig. 2, upper panel).

The consequence is that the high-SNR regime is systematically underfitted, and *any* mechanism that improves high-SNR behaviour will appear to add information. A structural prior is exactly such a mechanism.

C. The protocol

We propose the following control, which is cheap and which we recommend be run before any structural prior is credited.

- 1) Measure the per-bin gradient share of the training loss. If it is far from uniform, the regime receiving least gradient is underfitted.
- 2) Retrain *both* the structured and unstructured arms with training concentrated on the regime of interest, matching data budget, schedule, seed and initialization.
- 3) Re-measure the contrast. What survives is attributable to the prior; what does not was compensating for training inadequacy.

[STUB — RESULT OF APPLYING THE PROTOCOL]

The paragraph reporting what the control did to Δ_H , and its figure. Waiting on batch C2, which measures the seed spread of the mixed-SNR contrast; the focused contrast is already measured on three seeds. No scalar may be quoted here until then, and the figure must be rebuilt from C2 and C3 rather than repurposed — the retired one plotted a single seed of each design as though the pair were a measurement.

Scope of this negative result. It holds at $N = 32$, $K = 3$, this data budget, and the two pilot counts tested. It is not a general claim that structural priors cannot help unrolled estimators. Indeed the same prior applied to the *classical* estimator, where no training adequacy is at stake, gives a

TABLE I
CONFIGURATION.

Parameter	Value
Array elements N / users K / pilots P	32 / 3 / 20
Paths per user L_k	$\mathcal{U}\{3, 7\}$ [6]
SNR (train and test)	$\mathcal{U}[-10, 20]$ dB
Reference-to-signal ratio	10 dB
Unrolled layers T	10
Trainable parameters	1,586,900
Training samples / epochs	80,000 / 13
Test realizations	2,000 (paired)

large and stable gain, and under a distribution shift in path richness the structured network degrades least (+0.48 dB against +0.73 dB). The prior buys robustness here; it is the in-distribution accuracy claim that does not survive the control.

IV. NUMERICAL RESULTS

Table I lists the configuration. The reported statistic is the paired per-trial median within an SNR bin, with bootstrap 95% confidence intervals over 2000 paired test realizations.

A. The known remedy transfers to this setting

The confound is a property of the loss, and the remedy for it is already known [5]. This subsection reports what it is worth here. That is a confirmation, not a contribution, and it is placed first only because the rest of the section is easier to read once the loss is fixed.

We weight each sample by a static per-bin factor $w(b) = c/m(b)$, where $m(b)$ is the mean per-sample normalized error in bin b measured once before training, with c set so the weights have unit mean. This requires no schedule, no tuning, and no change to the architecture.

It works as designed. The realized per-bin gradient share spans a factor of 1.9 after balancing against 30.0 before, and the share below 5 dB falls from 0.897 to 0.427 against an ideal of 0.5 — a slight over-correction.

The estimation result is better than we predicted, and it is measured on three seeds rather than one. Against identically-trained unbalanced baselines, balancing gains +1.902, +2.242 and +2.369 dB at $\text{SNR} \geq 5$ dB, a mean of +2.171 dB. Each seed's bootstrap 95% confidence interval over test realizations excludes zero. The spread over seeds is a separate quantity and is reported as one: its standard deviation is 0.241 dB, on three seeds and therefore two degrees of freedom, so it is indicative rather than precise. Taking the conservative combination, mean $- 2 \text{ SD} = +1.689$ dB, the effect is separated from zero across seeds.

We had pre-registered a low-SNR cost of -0.10 to -0.40 dB on the reasoning that the lowest bin's weight falls about eighteenfold. There was no cost. On the seed for which the prediction was registered the low-SNR change is +0.135 dB, CI over test realizations [+0.101, +0.172]; across the three seeds the two lowest bins gain +0.052 and +0.202 dB on average. Downweighting the bins that dominated the gradient did not harm them, because they were over-served rather than merely well-served.

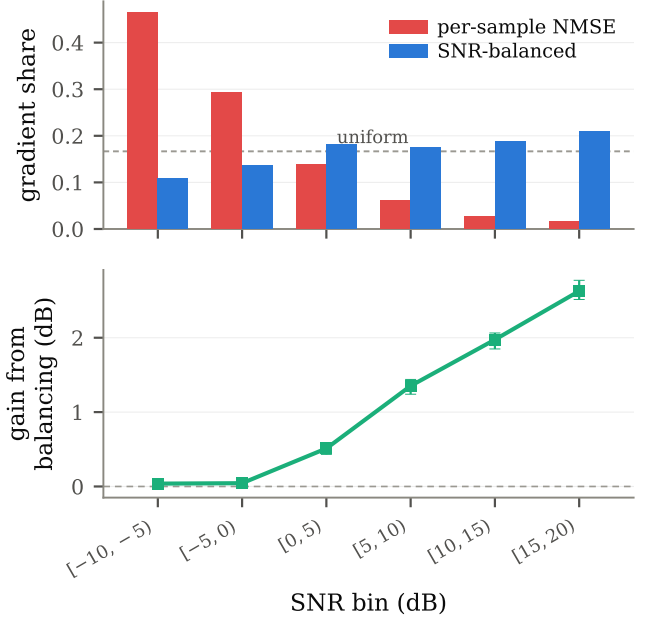


Fig. 2. Cause and effect. Upper: per-bin gradient share under the per-sample normalized loss and under SNR balancing. Lower: the resulting per-bin gain, paired per-trial median with bootstrap 95% CI over test realizations, on the seed for which the prediction was registered.

The gap to the genie-aided reference at $\text{SNR} \geq 5$ dB falls from +2.897 to +1.000 dB. Below 5 dB the balanced estimator passes that reference by up to 1.99 dB, which is consistent rather than anomalous: the reference is restricted to unstructured least squares, and an estimator that has learned the channel geometry is not so restricted.

B. The correction improves the regime it penalizes

The obvious objection to a reweighting is that it trades one regime for another: whatever the high-SNR bins gain, the down-weighted low-SNR bins must have paid. Here they did not, and the check is exhaustive rather than pooled.

Fig. 3 plots every SNR bin for every seed. **All eighteen bin-by-seed cells favour the balanced loss**, and seventeen of the eighteen have a bootstrap 95% confidence interval over test realizations that excludes zero; the single exception is seed 1's $[-5, 0)$ dB bin at +0.044 dB, CI $[-0.011, +0.105]$. The three-seed means rise monotonically across the range, from +0.052 dB in $[-10, -5)$ to +3.120 dB in $[15, 20)$.

The two lowest bins are the ones that matter for the objection, because the weighting divides their contribution by about 18 and 9 relative to unit mean. They gain anyway. The correction is therefore not a transfer between regimes: it improves the quantity it was weighted away from.

We offer a reading of why, as a hypothesis and not as a result. Under the per-sample loss the gradient was so dominated by the low-SNR tail that the network was underfitted everywhere, spending capacity on realizations from which little is recoverable. Rebalancing does not move performance from one regime to another; it stops wasting it. Testing that claim directly would require an ablation we have not run.

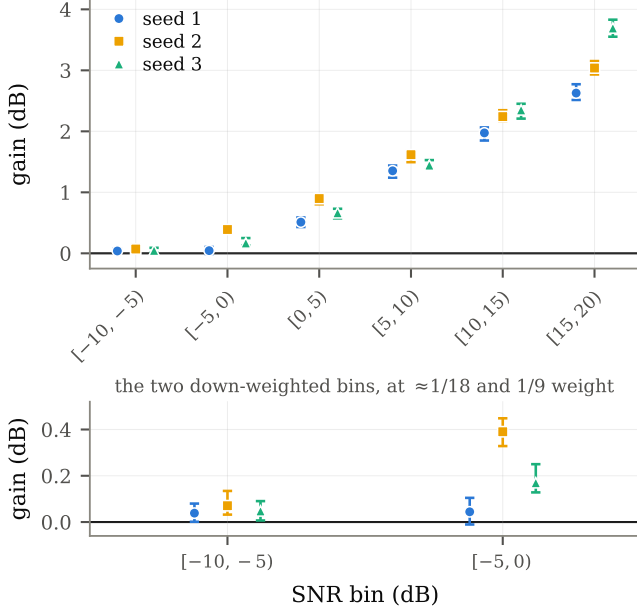


Fig. 3. Every bin, every seed. The gain from SNR balancing, paired per-trial median with bootstrap 95% CI over test realizations, for each of three training seeds. All eighteen cells are positive, including the two lowest bins, whose weights the balancing divides by about 18 and 9.

TABLE II
PER-BIN STRUCTURAL GAIN Δ_H , THREE-SEED MEAN, POSITIVE = THE PRIOR IS BETTER. EVERY ONE OF THE EIGHTEEN UNDERLYING BIN-BY-SEED CELLS IN THE MIXED DESIGN HAS A BOOTSTRAP 95% CI OVER TEST REALIZATIONS EXCLUDING ZERO.

SNR bin (dB)	mixed	SD over seeds	focused
$[-10, -5)$	-0.120	0.036	—
$[-5, 0)$	-0.400	0.103	—
$[0, 5)$	-0.293	0.157	—
$[5, 10)$	+0.457	0.052	-0.173
$[10, 15)$	+1.403	0.089	+0.067
$[15, 20)$	+2.290	0.082	+0.259

C. The structural contrast is not a scalar: a per-bin sign pattern

Reported as one number, the structural gain Δ_H is a weighted average of two opposing effects. Per bin, the prior costs accuracy at low SNR and buys it at high SNR, and the sign flips exactly once.

Two things about Table II are worth separating.

The ordering replicates and the crossing does not. Every arm we have measured — three seeds of mixed-SNR training, three of focused training, and an earlier arm under the balanced loss — places every negative bin before every positive one. On all three mixed seeds the sequence is exactly — — — + + + with all eighteen intervals excluding zero. But the bin at which the sign turns moves by up to two positions between arms. The ordering is the property that survives; the crossing SNR is not, and neither is any scalar summary that depends on where it falls.

The magnitude collapses under matched adequacy while the ordering does not. Comparing the two columns, focused

training leaves the shape intact and reduces the size of every bin by roughly an order of magnitude.

D. The averaging window decides the reported sign

Because the per-bin contributions have opposite signs, the number one reports depends on the window the average is taken over — both which bins are inside it and how trials are combined within it. Across three seeds of each design:

- mixed-SNR, restricted to $\text{SNR} \geq 5$: +1.209, +1.351, +1.298 dB by paired per-trial median; +0.723, +0.856, +0.901 dB by ratio of sums. Positive under both.
- mixed-SNR, pooled over the whole range: +0.129, +0.225, +0.196 dB by median; -0.214, -0.146, -0.141 dB by ratio of sums. *Sign disagreement*.
- focused training, $\text{SNR} \geq 5$: +0.078, +0.057, -0.076 dB by median; -0.059, -0.077, -0.234 dB by ratio of sums. Negative under ratio of sums on every seed.

The reason is measurable and we measured it. The paired median weights every trial equally; the ratio of sums weights each trial by its own squared error. Sorting trials by the no-prior arm's error, the top two deciles carry 51.2% of the total and contribute -0.134 dB, while the remaining eight contribute essentially nothing. That is not a conditioning effect separate from SNR: within an SNR bin the rank correlation between the baseline error and Δ_H is -0.009, -0.046 and +0.147, against -0.161 overall and +0.198 against SNR itself. The sum of squared errors is dominated by low-SNR realizations, which is exactly where Table II says the prior costs accuracy.

It is not the mechanism that produces the same symptom in the classical estimator, where a selection rule abstains on most trials and pins the median at zero. The unrolled arm has no such rule: the projection is applied unconditionally, and it moves the estimate on every one of the 10 layers of every one of 2000 trials on every seed, with no trial anywhere near inactive.

E. The structural prior under matched adequacy

[STUB — STRUCTURAL-PRIOR SECTION]

The headline comparison and its figure, pending batch C3. Its per-bin and averaging-window content has been promoted out into Secs. IV-C and IV-D; what remains to be written here is the scalar contrast across training designs and the claim the paper makes from it. The measured inputs are: $\Delta_H = +1.286$ dB under mixed-SNR training, across-seed SD 0.072, mean - 2 SD = +1.142, positive on every seed and under both pooling rules; against +0.020 dB under matched focused training, SD 0.083, mean - 2 SD = -0.147, one seed of three significantly negative. The claim is that the contrast changes category rather than shrinking by a factor — from an effect that survives every check to one whose sign depends on the seed drawn and the statistic used. C3 tests whether a third way of removing the training imbalance, reweighting the loss without narrowing the training range, does the same thing; the section is not written until it reports, because a claim about training adequacy in general rests on more than one way of fixing it.

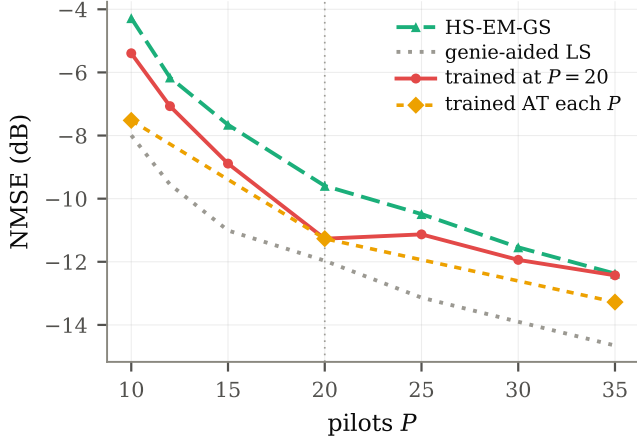


Fig. 4. Pilot efficiency against pilot-count generalization at 5 dB, on identical realizations. The dashed diamonds are separate networks trained at each P ; the solid curve is one network trained at $P = 20$.

F. Pilot efficiency is not pilot-count generalization

A pilot-count curve for a trained estimator can mean two different things: one model trained at some P_0 and evaluated at each P (generalization), or a model trained at each P (efficiency). These are not close. Trained at $P = 10$, the estimator beats the $P = 20$ model evaluated there by +2.231 dB pooled, every bin’s CI excluding zero. At $P = 35$ the corresponding margin is +0.635 dB, so the model generalizes upward in pilot count far better than downward.

At 5 dB the matched-trained estimator reaches -7.52 dB at $P = 10$, within 0.47 dB of the genie-aided reference, while the $P = 20$ model evaluated there is 2.60 dB short of it. The clearest single indication that the curve is tied to its training condition is that the $P = 20$ model is *worse* at $P = 25$ (-11.13 dB) than at $P = 20$ (-11.27 dB): more measurements, worse estimate. Fig. 4 shows all three curves together.

This matters for how such curves are reported. The pilot sweep of [3] is described as “the NMSE performance versus the number of pilot transmissions P , evaluated at a fixed SNR of 5 dB”, and as “crucial for assessing the pilot efficiency of each algorithm”; the text does not state whether the networks were retrained per pilot count. We report that as a fact about what the paper specifies, and as motivation for stating the distinction explicitly — not as a criticism of that work, whose convention is the ordinary one and which our own earlier curves shared.

V. CONCLUSION

A training loss averaged over a wide SNR range is dominated by its low-SNR tail [5]; here that amounts to 89.7% of the gradient below 5 dB and leaves the high-SNR regime underfitted. The consequence we draw from it is about evidence rather than about performance: any mechanism that improves high-SNR behaviour will appear to add information it need not have supplied, so a structural prior credited on an uncontrolled comparison is credited for the training’s failure.

We recommend the matched-adequacy control be run before a structural prior is credited, and note that it cost us our own result first.

Two properties of the structural contrast survive that control and a single number does not report either: the per-bin sign pattern, which replicates across training designs and seeds while the SNR at which it crosses does not, and the dependence of the reported sign on the averaging window. Applying the known remedy [5] to this setting gains +2.171 dB at $\text{SNR} \geq 5$ dB across three seeds and improves every one of eighteen bin-by-seed cells, including the two the weighting itself penalizes most.

[STUB — CONCLUSION, structural-prior sentence]

One or two sentences stating what the control did to Δ_H , pending C2. The claim the retired draft made here — a fifteenfold collapse — is withdrawn and must not return in any form.

Limitations. All results are simulation only, with no measured hardware and no recorded channel data. They cover the geometric ULA model of (1) at $N = 32$, $K = 3$, one data budget and the pilot counts tested; the negative result about structural priors is scoped to those conditions and is not a general claim. We establish no fundamental bound: the reference curve used throughout is genie-aided, presumes phase no receiver observes, and is passed by our own estimator at low SNR.

REFERENCES

- [1] K. Gregor and Y. LeCun, “Learning fast approximations of sparse coding,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 399–406.
- [2] V. Monga, Y. Li, and Y. C. Eldar, “Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing,” *IEEE Signal Process. Mag.*, vol. 38, no. 2, pp. 18–44, 2021.
- [3] J. Xiao, J. Wang, M. Zeng, H. Xu, X. Li, and A. Nallanathan, “Channel estimation for Rydberg atomic quantum receivers: Unrolled phase retrieval from holographic snapshots,” *IEEE Signal Process. Lett.*, vol. 33, pp. 1696–1700, 2026.
- [4] M. Cui, Q. Zeng, and K. Huang, “Towards atomic MIMO receivers,” *IEEE J. Sel. Areas Commun.*, vol. 43, no. 3, pp. 659–673, 2025.
- [5] R. Wiesmayr, G. Marti, C. Dick, H. Song, and C. Studer, “Bit error and block error rate training for ML-assisted communication,” *arXiv preprint arXiv:2210.14103*, 2022, v3, Mar. 2023.
- [6] B. Xu, J. Zhang, Z. Chen, B. Cheng, Z. Liu, Y.-C. Wu, and B. Ai, “Channel estimation for Rydberg atomic receivers,” *IEEE Wireless Commun. Lett.*, vol. 14, no. 9, pp. 2957–2961, 2025, arXiv:2503.08985.
- [7] R. W. Gerchberg and W. O. Saxton, “A practical algorithm for the determination of phase from image and diffraction plane pictures,” *Optik*, vol. 35, pp. 237–246, 1972.